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Potential as "intrinsic energy of inertia"

Electro-Magnetic Field (force caused by potential)

- static: lovslized source charge at rest, only spacial dependency, no time dependency
- stationary: source charce at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistc: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency

- slow vs. fast movemnt
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m = \frac{1}{c}\left(\frac{r}{|r|} \times \mathcal{E}\right) \overset{\text{Far Field}}{=} \frac{1}{2}\frac{1}{2\pi}\mu_0 \frac{k^2 e^{i(kr-\omega t)}}{r}\frac{r}{|r|} \times \left(cp_{0_{dipol}} + m_{0_{dipol}} \times \frac{r}{|r|}\right) \overset{\text{Near Field}}{\approx} \frac{1}{2}\frac{1}{2\pi}\mu_0 q \frac{v \times r}{|r|^2} \tag{1}$$

Magnetic Field in Matter

Exitation, Displacement

$$\mathcal{H} = \ldots \tag{2}$$

Electric Field in Vacuum

$$\mathcal{E} \overset{\text{conservativ}}{=} -\nabla U_e \overset{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t}U_m \overset{\text{fast moving charge}}{=} \frac{1}{2}\frac{1}{2\pi}\frac{1}{\varepsilon_0}\frac{1}{r}\frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right)\frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} \ldots\right.\right.}{\ldots}}_{\text{accelerate}\ldots} \tag{3}$$

Electric Field in Matter

Exitation, Displacement (field change)
dielectricum = non-conducting matter
electricum = conductor medium

$$\mathcal{D} = \varepsilon_0\mathcal{E} + \mathcal{P} = \overset{\text{isotropic medium in weak fields}}{=} \varepsilon_0\mathcal{E} + \varepsilon_0\chi\mathcal{E} = \varepsilon_0\left(1+\chi\right)\mathcal{E} = 4\pi r^3 n\varepsilon_0 \left\langle p_m\right\rangle = 4\pi r^3 n\varepsilon_0 \left\langle \mathcal{E}_m\right\rangle = \alpha n\varepsilon_{\ldots} \tag{4}$$

Abbreviations

$$\mathcal{E} = \text{Electric Field} \qquad \mathcal{B} = \text{Magnetic Fiels} \qquad \mathcal{D} = \text{Displacement density of dielectricum}$$

$$\mathcal{P} = \text{Polarisation} \qquad \mathcal{H} = \text{magn. exitation, magn. field strength ("magn. displacement")}$$

$$\mathcal{M} = \text{magnetisation density} \qquad n = \text{spacial molecule density in volume} \alpha = 4\pi r^3 = constant$$

$$\chi = \text{electic susceptibility of dieelectricum} \qquad \kappa = \frac{\mu - \mu_0}{\mu_0} = \text{magn. susceptibility}$$

$$\langle p_m \rangle = \text{spacial average magn. dipol moment} \qquad \langle \mathcal{E}_m \rangle = \text{spacial average of electric field}$$

$$\bar{\mathcal{E}}_m = \text{time average of electric field} \qquad \mu = \text{magn. permeability} \qquad \mu_0 = \text{magn. constant ...}$$

$$\varepsilon = \text{dielectricity constant} \qquad \varepsilon_0 = \text{...}$$

Electro-Magnetic interactions/coupling (Maxwell conditions)

conservativ, static charge localisation:

...

stationary, slow charge movement:

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \qquad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t}\mathcal{B} = i\omega\mathcal{B} \qquad \nabla \times \frac{1}{\mu_0}\mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t}\mathcal{E} = j - i\omega\varepsilon_0\mathcal{E} \tag{5}$$

relativistic, fast charge movement:

...

Potential

$\Rightarrow$ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}U = J \tag{6}$$

$\Rightarrow$ Potential caused by "ignition J" (inhomgenity) from source at rest, v = 0:

$$U = -\frac{1}{4\pi}\int \frac{J}{|r|}\, dr' \tag{7}$$

elec. Potential structure:

$J = -\frac{1}{\varepsilon_0}\varrho$

$$U_e := \Phi \overset{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi}\frac{1}{\varepsilon_0}\int \frac{\varrho}{|r|}\, dr' \mathrel{\hat{=}} \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\varepsilon_0}Q = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\varepsilon_0}It \tag{8}$$

magn. Potential structure:

$J = -\mu_0 j$

$$U_m := A \overset{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi}\int \frac{j}{|r|}\, dr' \mathrel{\hat{=}} \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\mu_0 I = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{c^2\varepsilon_0}I = \frac{1}{c^2}\frac{1}{t}U_e \overset{\text{source in uniformal movement},v=const}{=} \tag{9}$$

Electric endogen energy

- Bond Energy
- Binding Energy

$E_{pot} = mc^2$

$$e = \text{electron charge} \qquad q = ne = \text{charge} \qquad n \in \mathbb{N} \tag{10}$$

$$\text{Potential} \qquad U = \frac{E_{pot}}{q} = \frac{\left(mc^2\right)^2}{ne} = \frac{m}{q}mc^4 = \kappa mc^4 \tag{11}$$

Electric induced energy (exogenous excitation)

- Ionisation Energy
- Exitation Energy

$E_{kin} = \frac{1}{2}mv^2 \mathrel{\hat{=}} (pc)^2 \mathrel{\hat{=}} h\nu$

E_{kin} Energy density electric:

$$w_e = \frac{1}{2}\varepsilon_0\mathcal{E}^2 \tag{12}$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2 \tag{13}$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2}CU^2 \tag{14}$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2}LI^2 \tag{15}$$

Power:

$$P = VI = (U_1 - U_2)\frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \tag{16}$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \tag{17}$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c}\frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \tag{18}$$

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t}(we + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E}\right) dr = \frac{d}{dt}W_e + \frac{d}{dt}W_m + D_S + \int (j \cdot \mathcal{E})\, dr \overset{\text{low frequency}}{=} \frac{1}{2}\frac{d}{dt}(LI^2) + \frac{1}{2}\frac{d}{dt}\left(\frac{Q^2}{C}\right) \tag{19}$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \tag{20}$$

$$\Delta\cdot := \text{ div grad} = \text{Laplacian}$$

$$J = \text{"ignition of inhomogenity,cause"} \quad j = \text{Current density} \quad \varrho = \text{Charge density}$$

$$V = \text{Potentialdifference} \quad I = \text{Current} \quad R = \text{Resistance}$$

$$L = \text{Inductance} \qquad C = \text{Capacitance} \qquad Q = \text{charge (ion)} \qquad \Psi = LI = \text{mag. Flux}$$

$$\mathcal{S} = \mathcal{E} \times \mathcal{H} = \text{Poynting-Vector} = \text{radiation intensity}$$

$$D_S = \int (\,\cdot\,\mathcal{S})\; dr = \int \text{div}(\mathcal{E}\times\mathcal{H})\, dr = \text{dissipation engergy lost from radiation} \approx 0 \text{ for low frequencies } \omega$$

$$d = \sqrt{\frac{1}{\mu\sigma}\frac{2}{\omega}} = \text{skin depth (surface) of conductor} \qquad r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"}$$

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda\frac{1}{T} = \sqrt{\frac{k}{m}} \mathrel{\hat{=}} kc \mathrel{\hat{=}} \frac{1}{\sqrt{LC}} \mathrel{\hat{=}} \sqrt{\frac{g}{r}} = \text{frequency}$$

$$\mu = \text{permeability of conductor} \qquad \sigma = \text{conductivity of conductor (dielectricum)}$$

Low Frequency

$$\omega \ll \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \tag{21}$$

$$L \approx const. \qquad C \approx const. \qquad D_S \approx 0$$

High Frequency

$$\omega \gg \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \tag{22}$$

$$L(t) \neq const. \qquad C(t) \neq const. \qquad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Continuity at Rest: Electic Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emgergence of Radiation

Continuity and periodic (harmoneous oscillation) Movement in VACUUM

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \qquad \nabla \cdot j = i\omega\varrho \qquad i\omega Q = 0$$

Continuity and periodic (harmoneous oscillation) Movement in MATTER

Refraction index emerges

Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

$$r \ll \lambda = \frac{2\pi}{k} \overset{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \qquad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\varepsilon_0 r^6}}{\frac{k^2\mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$$\vec{p} = \text{elec. Dipol}$$

Far Field

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \overset{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \qquad kr \gg 1 \qquad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2}{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2} = 1$$

Dissipation in Vacuum

no Contiunuity of chrge/matter, no Refraction index, Reflection

Dissipation in Matter

no Contiunuity of chrge/matter with Dielectricum, Refraction merges, Reflection, Absorbtion

Energy gain by excitation:

Action Quantum, Planck Constant (Energy-Frequency-Slope)[1]:

$$h = \tan(\alpha) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \tag{23}$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \tag{24}$$

$$E_{kin}^{max} = eV_0 = e\frac{Q}{C} = h(\nu - \nu_0) \tag{25}$$

$$V_0 = \text{Opposing Potential } (I = 0)$$

$$\alpha = \angle(E_{kin}, \nu) \qquad e = \text{electron charge} \qquad Q = \text{ion charge}$$

$$C = \text{Capacitance} \qquad \nu = \frac{1}{2\pi}\frac{1}{\sqrt{LC}}\sqrt{1 - \frac{C}{L}R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C}\left(\frac{e}{h}Q + \nu_0\right)^2$$

$\Rightarrow$ Ionisation Energy of chemical element $E_i = h\nu = h\frac{\lambda}{c} = \hbar\omega = \frac{h}{k\lambda}\omega$

$\Rightarrow$ Energy released (gained) $E_{gain} = E_{pot} - E_i = mc^2 - h\nu$

[1] Wirkungsquantum